

Revisiting Stochastic Approximation and Stochastic Gradient Descent

Rajeeva Laxman Karandikar, Bhamidi Visweswara Rao and Mathukumalli Vidyasagar *

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Abstract

In this paper, we take a fresh look at stochastic approximation (SA) and Stochastic Gradient Descent (SGD). We derive new sufficient conditions for the convergence of SA. In particular, the “noise” or measurement error *need not* have a finite second moment, and under suitable conditions, not even a finite mean. By adapting this method of proof, we also derive sufficient conditions for the convergence of zero-order SGD, wherein the stochastic gradient is computed using *only two* function evaluations, and no gradient computations. The sufficient conditions derived here are the weakest to date, thus leading to a considerable expansion of the applicability of SA and SGD theory.

1 Introduction

Let $F : \mathbb{R} \mapsto \mathbb{R}$ be a twice continuously differentiable convex function such that $\exists a, b \in (0, \infty)$ satisfying

$$a \leq \frac{d^2}{dx^2} F(x) \leq b, \quad \forall x \in \mathbb{R}. \quad (1)$$

Our interest is in finding the (unique) x^* such that $F(x^*) \leq F(x)$ for all $x \in \mathbb{R}$. Starting with any x_0 , if we define for $n \geq 0$, writing $G(x) = \frac{d}{dx} F(x)$,

$$x_{n+1} = x_n - \frac{1}{b} G(x_n)$$

it can be shown that x_n converges to x^* . If $G(x)$ is not directly observable, but can be observed in presence of noise, then we can define $X_0 = x_0$ and for $n \geq 0$ let

$$X_{n+1} = X_n - \frac{\beta_n}{b} (G(X_n) + W_{n+1}) \quad (2)$$

where W_{n+1} is the observational error or noise, and $\beta_n \in (0, 1]$. When $\{W_n : n \geq 1\}$ are i.i.d. random variables with mean zero and finite variance, it can be shown that X_n converges to x^* , if $\{\beta_n : n \geq 0\}$ satisfy

$$\sum_n \beta_n = \infty, \quad \sum_n \beta_n^2 < \infty. \quad (3)$$

Robbins and Monro in 1951 [22] proved convergence of X_n to x^* in probability in a statistical context. Indeed, the conditions (3) are known as Robbins-Monro conditions.

*RLK and BVR are with the Chennai Mathematical Institute; MV is with the Indian Institute of Technology, Hyderabad. Emails: rlk@cmi.ac.in, bvrao@cmi.ac.in, m.vidyasagar@iith.ac.in

If $G(x) = \frac{d}{dx}F(x)$ is not observable but only $F(x)$ can be observed in presence of noise, one can define approximating sequence by $Y_0 = y_0$ and for $n \geq 0$ and $\eta_n, c_n \in (0, 1)$

$$Y_{n+1} = Y_n - \frac{\eta_n}{c_n} ((F(Y_n + c_n) + M'_{n+1}) - (F(Y_n - c_n) + M''_{n+1})). \quad (4)$$

When the errors $\{M'_{n+1} : n \geq 0\}$, $\{M''_{n+1} : n \geq 0\}$ are independent copies of i.i.d. mean zero random variables with finite variance, Kiefer and Wolfowitz [15] had shown (again in a statistical context) that Y_n also converges to x^* in probability under suitable conditions on $\eta_n, c_n : n \geq 0$. This has been named Stochastic Gradient Descent.

Blum [3], [4] showed that convergence of X_n (defined by (2)) and Y_n (defined by (4)) to x^* is almost sure, and also proved the d -dimensional versions of these results.

There has been lot of expansions of these themes, named Stochastic Approximation and Stochastic Gradient Descent. Here are some references on these themes : [1], [2], [6], [7], [8], [11], [12], [14], [16], [17], [18], [19], [20], [21], [23], [24], [25].

Some of these approaches rely on Martingale techniques (which in turn relies on various L^2 estimates of the approximating sequence) while some embed the problem to a differential equation in continuous time.

The main contribution of this paper is to separately analyse the behaviour of the noise sequence in (1) and (2) and show that if the moment conditions on noise and suitable choice of step size sequences (β_n in (1) and η_n in (2) are suitably chosen along with c_n) are such that the noise sequence satisfies a version of law of large numbers, then Stochastic Approximation and Stochastic Gradient Descent are also valid (under appropriate conditions on the functions G and F). We are able to show that if errors are i.i.d. with finite first moment (and mean 0), then Stochastic Approximation holds with $\beta_n = \frac{1}{n}$ and Stochastic Gradient Descent is valid with $\eta_n = \frac{1}{n \log(1+n)}$, $c_n = \frac{1}{\log(1+n)}$. The proofs in these cases follow steps involved in Kolmogorov's proof of Strong law of large numbers in the case of i.i.d. L^1 random variables. See [9], [10], [13].

We will also show that, if errors form a martingale difference sequence that is L^α bounded, $\alpha > 1$, then Stochastic Approximation holds with $\beta_n = \frac{1}{n}$ and Stochastic Gradient Descent is valid with $\eta_n = \frac{1}{n \log(1+n)}$, $c_n = \frac{1}{\log(1+n)}$.

2 Stochastic Approximation - Main Result

Let $d \geq 1$ be a fixed integer and $G : \mathbb{R}^d \mapsto \mathbb{R}^d$ be a function with a unique zero: *i.e.* $\exists x^*$ such that $G(x^*) = 0$ and $G(x) \neq 0$ for all $x \neq x^*$. We are going to consider d -dimensional version of (2) in which the errors may be inflated by the history (at most linearly). So we consider the following setup: For $n \geq 0$, let $\lambda_n : (\mathbb{R}^d)^{n+1} \mapsto \mathbb{R}$ be such that for some constant C_1 one has

$$|\lambda_n(u_0, u_1, \dots, u_n)| \leq C_1(1 + \max_{0 \leq k \leq n} \|u_k\|). \quad (5)$$

For $x_0 \in \mathbb{R}^d$ fixed, $X_0 = x_0$ and for $n \geq 0$ let

$$X_{n+1} = X_n - \beta_n(G(X_n) + \lambda_n(X_0, \dots, X_n)W_{n+1}). \quad (6)$$

Here and in what follows, $\|x\|$ is the Euclidian norm (l^2 -norm) on $\mathbb{R}^d$. We are going to assume that the function $G : \mathbb{R}^d \mapsto \mathbb{R}^d$ satisfies

$$\exists b > 0, \rho < 1, x^* \in \mathbb{R}^d \text{ such that } \|x - x^* - \frac{1}{b}G(x)\| \leq \rho\|x - x^*\| \quad \forall x \in \mathbb{R}^d. \quad (7)$$

and going to look for conditions on the *step size* sequence $\{\beta_n : n \geq 0\}$ and noise sequence $\{W_n : n \geq 1\}$ that ensure convergence of $\{X_n : n \geq 0\}$ defined by (6) to x^* . On $\mathbb{R}^d$, all norms are equivalent and we can consider (7) with l^∞ norm as well. When $G(x) = \frac{d}{dx}F(x)$ with F satisfying (1), then it can be seen that (7) holds. It may be noted that the constants b and ρ appearing in the condition (7) are not needed to construct and implement the algorithm. These appear only in the proofs.

Let us take the special case of this - when $G(x) = tx$, $t > 0$ and $\{\zeta_n : n \geq 0\}$ are bounded $\mathcal{F}_n = \sigma(X_0, \dots, X_n)$ adapted random variables. Let $U_0 = 0$ and for $n \geq 0$, let

$$U_{n+1} = (1 - t\beta_n)U_n + t\beta_n\zeta_n W_{n+1}, \quad n \geq 0.$$

When $\beta_n = \frac{1}{n+1}$, $t = 1$ and $\zeta_n = 1$ for all $n \geq 0$, then $U_n = \frac{W_1 + W_2 + \dots + W_n}{n}$ and thus almost sure convergence of U_n to 0 is same as $\{W_n : n \geq 1\}$ satisfying the strong law of large numbers (SLLN) (with mean 0). It can be seen that when $\{W_n : n \geq 0\}$ are i.i.d. with mean 0 and finite variance, to conclude that U_n converges to 0, we must have

$$\lim_{n \rightarrow \infty} \beta_n = 0, \quad \sum_{n=0}^{\infty} \beta_n = \infty. \quad (8)$$

This leads to the following definition: Let $(\mathcal{F}_n)_{n \geq 0}$ be a filtration on a probability space $(\Omega, \mathcal{F}, \mathbb{P})$.

Definition : Let $\{\beta_n : n \geq 0\}$ satisfy (8). An $\mathbb{R}^d$ valued $(\mathcal{F}_n)_{n \geq 0}$ adapted sequence of Random variables $\{W_n : n \geq 1\}$ is said to satisfy **Generalised Strong Law of Large Numbers (GSLLN)** at the rate $\{\beta_n : n \geq 0\}$ w.r.t. the filtration $(\mathcal{F}_n)_{n \geq 0}$ if for all uniformly bounded $(\mathcal{F}_n)_{n \geq 0}$ adapted random variables $\{\zeta_n : n \geq 0\}$ and for all $t \in (0, \infty)$ the sequence $\{S_n : n \geq 0\}$ defined by $S_0 = 0$ and

$$S_{n+1} = (1 - t\beta_n)S_n + t\beta_n\zeta_n W_{n+1}, \quad n \geq 0 \quad (9)$$

converges to 0 almost surely.

If there is a fixed filtration in consideration, we will drop it from the statement and say it as $\{W_n : n \geq 1\}$ satisfies GSLLN at the rate $\{\beta_n : n \geq 0\}$. Clearly, the set of $(\mathcal{F}_n)_{n \geq 0}$ adapted sequences of random variables that satisfy GSLLN at the rate $\{\beta_n : n \geq 0\}$ w.r.t. the filtration $(\mathcal{F}_n)_{n \geq 0}$ is a linear space.

Given a sequence of random variables $\{W_n : n \geq 1\}$ on $(\Omega, \mathcal{F}, \mathbb{P})$, let $(\mathcal{F}^W)$ denote the filtration $\mathcal{F}_0^W = \{\Phi, \Omega\}$, and for $n \geq 1$, $\mathcal{F}_n^W = \sigma(W_1, \dots, W_n)$.

Our main result on **Stochastic Approximation** is the following:

THEOREM 1. Suppose $G : \mathbb{R}^d \mapsto \mathbb{R}^d$ satisfy (7). Let $\lambda_n : (\mathbb{R}^d)^{n+1} \mapsto \mathbb{R} : n \geq 0$ satisfy (5). Let $\{\beta_n : n \geq 0\}$ satisfy (8) and let $\{W_n : n \geq 1\}$ be a sequence of $\mathbb{R}^d$ valued random variables. Suppose $\{W_n : n \geq 1\}$ satisfies Generalised Strong Law of Large Numbers at the rate $\{\beta_n : n \geq 0\}$ w.r.t. the filtration $(\mathcal{F}^W)$. Let $x_0 \in \mathbb{R}^d$. Let $\{X_n : n \geq 0\}$ be defined by (6). Then

$$\|X_{n+1} - x^*\| \rightarrow 0 \text{ almost surely.} \quad (10)$$

It should be noted that conditions on G and those on the noise sequence $\{W_n : n \geq 1\}$ are not related to each other. Of course, the conditions on the rate and the errors are related.

3 Stochastic Gradient Descent - Main Result

In this section, we assume that $F : \mathbb{R}^d \mapsto \mathbb{R}$ admits a unique minimum $\exists x^* = (x_1^*, x_2^*, \dots, x_d^*) \in \mathbb{R}^d$ (i.e. $F(x^*) < F(x)$ for all $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$, $x \neq x^*$). Here, $e_1, e_2, \dots, e_d$ will denote the standard orthonormal basis of $\mathbb{R}^d$. We will also assume that $\exists \rho \in (0, 1)$, $a > 0$, $b > 0$ such that $\forall c > 0$,

$$|x_i - x_i^* - \frac{1}{2cb}(F(x + ce_i) - F(x - ce_i))| \leq \rho|x_i - x_i^*| + ac, \quad \forall 1 \leq i \leq d. \quad (11)$$

Here, for $x \in \mathbb{R}^d$, $\|x\|_\infty$ denotes the l^∞ norm for $x = (x_1, x_2, \dots, x_d) \in \mathbb{R}^d$, $\|x\|_\infty = \max\{|x_i| : 1 \leq i \leq d\}$. In the rest of the paper, just like d is a fixed integer, i will denote an integer in $\{1, 2, \dots, d\}$. Statement such as for all n, i will mean for all integers $n \geq 0$ and $i \in \{1, 2, \dots, d\}$. Let $\lambda_n : (\mathbb{R}^d)^{n+1} \mapsto \mathbb{R} : n \geq 0$ be such that (for some constant C_1)

$$|\lambda_n(u_0, u_1, \dots, u_n)| \leq C_1(1 + \max_{0 \leq k \leq n} \|u_k\|_\infty). \quad (12)$$

Let $\{\beta_n : n \geq 0\} \subseteq (0, 1)$, $\{c_n : n \geq 0\} \subseteq (0, 1)$ satisfy

$$\lim_{n \rightarrow \infty} \eta_n = 0, \quad \lim_{n \rightarrow \infty} c_n = 0. \quad (13)$$

Let $\{M'_m, M''_m : m \geq 1\}$ be sequences of $\mathbb{R}^d$ valued random variables (the observational errors appearing in (15) below) Let $\{\psi_{n,i} : n \geq 0, 1 \leq i \leq d\}$ be $\{0, 1\}$ valued $(\mathcal{F}^{M', M''})$ adapted random variables. We will also assume that $\forall i \in \{1, 2, \dots, d\}$

$$\sum_{n=0}^{\infty} \psi_{n,i} \eta_n = \infty \quad \text{almost surely.} \quad (14)$$

For $y_0 \in \mathbb{R}^d$ fixed, $Y_0 = y_0$ and for $n \geq 0$ let $\xi_n = \lambda_n(Y_0, \dots, Y_n)$ and

$$Y_{n+1} = Y_n - \sum_i \psi_{n,i} \eta_n \left(\frac{(F(Y_n + e_i c_n) + \xi_n M'_{n+1,i}) - (F(Y_n - e_i c_n) + \xi_n M''_{n+1,i})}{2c_n} \right) e_i. \quad (15)$$

Our main result on **Stochastic Gradient Descent** is the following:

THEOREM 2. *Suppose $F : \mathbb{R}^d \mapsto \mathbb{R}$ satisfies (11). Let $\lambda_n : (\mathbb{R}^d)^{n+1} \mapsto \mathbb{R} : n \geq 0$ satisfy (12). Let $\{M'_n, M''_n : n \geq 1\}$ be $\mathbb{R}^d$ valued random variables. Let $\{\eta_n, c_n, \psi_{n,i} : n \geq 0, 1 \leq i \leq d\}$ satisfy (13), (14). Suppose $\psi_{\cdot,i}$ are adapted to $(\mathcal{F}^{M', M''})$.*

Let $\{Y_n : n \geq 0\}$ be defined by (15). Let $W_{n+1} = \frac{1}{2c_n}(M'_{n+1} - M''_{n+1})$ for $n \geq 0$.

Suppose the error sequences $\{W_n : n \geq 1\}$ satisfies Generalised Strong Law of Large Numbers at the rate $\{\eta_n : n \geq 0\}$ w.r.t. $(\mathcal{F}^{M', M''})$. Then

$$\|Y_{n+1} - x^*\| \rightarrow 0 \quad \text{almost surely.} \quad (16)$$

4 Generalised Strong Law of Large Numbers

We have seen that for both the questions: validity of Stochastic Approximation as well as that of Stochastic Gradient Descent (under suitable assumptions on F and G respectively) reduces to validity of Generalised Strong Law of Large Numbers for the noise (or error) sequences.

We will prove later that if $\{W_n : n \geq 1\}$ are i.i.d. with mean 0 and $\mathbb{E}[\|W_1\|^\alpha] < \infty$ for $1 \leq \alpha < 2$, then GSLLN holds if $\beta_n \leq Cn^{-\frac{1}{\alpha}}$. Of course, if $\mathbb{E}[\|W_1\|^2] < \infty$, then GSLLN are true if Robbins-Monro conditions hold. We will also have a version where GSLLN is true even if $\mathbb{E}[\|W_1\|] = \infty$. Moreover, we will also show that if $\{W_n : n \geq 1\}$ is a martingale difference sequence with $\sup_n \mathbb{E}[\|W_n\|^\alpha] < \infty$ for $1 < \alpha \leq 2$, then GSLLN holds if $\sum_n \beta_n^\alpha < \infty$.

Here is first of the two important results on GSLLN :

THEOREM 3. *Let $\{W_n : n \geq 1\}$ be an $\mathbb{R}^d$ valued $(\mathcal{F}_n)$ adapted sequence such that for each $n \geq 1$, $i \in \{1, 2, \dots, d\}$ $\exists A_{n,i} \in \mathcal{F}_n$ such that, writing*

$$V_{n+1,i} = \mathbb{E}[W_{n+1,i} \mathbf{1}_{A_{n+1,i}} \mid \mathcal{F}_n] \quad (17)$$

we have

$$\sum_{n=1}^{\infty} P(A_{n,i}^c) < \infty, \quad \forall i. \quad (18)$$

$$\sum_{n=0}^{\infty} \beta_n \mathbb{E}[|V_{n+1,i}|] \text{ converges } \forall i. \quad (19)$$

$$\sum_{n=0}^{\infty} \beta_n^2 \mathbb{E}[|W_{n+1,i}|^2 \mathbf{1}_{A_{n+1,i}}] < \infty. \quad (20)$$

Then

$$\sum_{n=0}^{\infty} \beta_n W_{n+1,i} \text{ converges } \forall i. \quad (21)$$

and hence $\{W_n : n \geq 1\}$ satisfies Generalised Strong Law of Large Numbers at the rate $\{\beta_n : n \geq 0\}$.

Here is the alternate version, that is very similar to the proof of Kolmogorov's Strong Law of Large numbers.

THEOREM 4. *Let $\{W_n : n \geq 1\}$ be an $\mathbb{R}^d$ valued $(\mathcal{F}_n)$ adapted sequence such that for each $n \geq 1$, $i \in \{1, 2, \dots, d\}$ $\exists A_{n,i} \in \mathcal{F}_n$ such that (18) and (20) hold and with $V_{n+1,i}$ defined by (17), we have*

$$\lim_{n \rightarrow \infty} V_{n+1,i} = 0, \quad \text{almost surely } \forall i \quad (22)$$

Then $\{W_n : n \geq 1\}$ satisfies Generalised Strong Law of Large Numbers at the rate $\{\beta_n : n \geq 0\}$.

Our next result lists several different conditions, any one of them ensures GSLLN holds, and as a consequence so does Stochastic Approximation.

THEOREM 5. *Suppose $\{\beta_n : n \geq 0\}$ and $\{W_n : n \geq 1\}$ satisfy any one of the conditions (H1)–(H5) given below (in addition to $\{\beta_n : n \geq 0\}$ satisfying (8)), then $\{W_n : n \geq 1\}$ satisfies Generalised Strong Law of Large Numbers at the rate $\{\beta_n : n \geq 0\}$. As a consequence, Theorem 1 also holds.*

(H1) $\{W_n : n \geq 1\}$ are i.i.d. with $\mathbb{E}(\|W_1\|^2) < \infty$, $\mathbb{E}(W_1) = 0$ and

$$\sum_{n=0}^{\infty} \beta_n^2 < \infty. \quad (23)$$

(H2) $\{W_n : n \geq 1\}$ are i.i.d. with $\mathbb{E}(\|W_1\|^\alpha) < \infty$, $1 \leq \alpha < 2$ and $\mathbb{E}(W_1) = 0$ and for some finite constant D ,

$$\beta_n \leq Dn^{-\frac{1}{\alpha}}, \quad \forall n \geq 1. \quad (24)$$

(H3) $\{W_n : n \geq 1\}$ are i.i.d. with symmetric distribution (i.e. W_1 and $-W_1$ have the same distribution), for some $0 < \delta \leq 1$, $\mathbb{E}[\frac{|W_{1,i}|}{(\log(1+|W_{1,i}|))^\delta}] < \infty$ for each i and for some finite constant D , $\beta_0 \leq D$ and

$$\beta_n \leq \frac{D}{n(\log(1+n))^\delta}, \quad \forall n \geq 1. \quad (25)$$

(H4) $\{W_n : n \geq 1\}$ are independent, $\mu_n = \mathbb{E}(W_n)$ converges to zero, for some α , $1 < \alpha \leq 2$, $\nu_{\alpha,n} = \mathbb{E}(\|W_n\|^\alpha) < \infty$ for all $n \geq 1$ and

$$\sum_{n=0}^{\infty} \beta_n^\alpha \nu_{\alpha,n+1} < \infty. \quad (26)$$

(H5) $\{W_n : n \geq 1\}$ is a martingale difference sequence, for some α , $1 < \alpha \leq 2$, $\nu_{\alpha,n} = \mathbb{E}(\|W_n\|^\alpha) < \infty$ for all $n \geq 1$ and

$$\sum_{n=0}^{\infty} \beta_n^\alpha \nu_{\alpha,n+1} < \infty. \quad (27)$$

Here are similar set of conditions for GSSLN that lead to Stochastic Gradient Descent

THEOREM 6. Let $\{M'_n, M''_n : n \geq 1\}$ be $\mathbb{R}^d$ valued random variables, let $\{\eta_n, c_n : n \geq 0\}$ satisfy (13). Let $M_n = \frac{1}{2}(M'_n - M''_n)$. Let $W_{n+1} = \frac{1}{c_n} M_{n+1}$.

Suppose that one of the conditions (K1)–(K5) given below holds. Then GSSLN holds for $\{W_{n+1} : n \geq 0\}$ at the rate $\{\eta_n : n \geq 0\}$, and as a consequence, Stochastic Gradient Descent Theorem 2 holds (if in addition, (11), (12), (13), (14) hold).

(K1) Suppose $\{M'_n, M''_n : n \geq 1\}$ are i.i.d. with $\mathbb{E}(\|M'_1\|^2) < \infty$ and

$$\sum_{n=0}^{\infty} \frac{\eta_n^2}{c_n^2} < \infty. \quad (28)$$

(K2) Suppose $\{M'_n, M''_n : n \geq 1\}$ are i.i.d., for some α , $1 \leq \alpha < 2$, $\mathbb{E}(\|M'_1\|^\alpha) < \infty$, and for some finite constant D , $\beta_0 \leq D$ and

$$\frac{\eta_n}{c_n} \leq Dn^{-\frac{1}{\alpha}}, \quad \forall n \geq 1. \quad (29)$$

(K3) Suppose $\{M'_n, M''_n : n \geq 1\}$ are i.i.d., for some $0 < \delta \leq 1$, $\mathbb{E}[\frac{|M'_{1,i}|}{(\log(1+|M'_{1,i}|))^\delta}] < \infty$ for each i and for some finite constant D , $\beta_0 \leq D$ and

$$\frac{\eta_n}{c_n} \leq \frac{D}{n(\log(1+n))^\delta}, \quad \forall n \geq 1. \quad (30)$$

(K4) $\{M_n : n \geq 1\}$ are independent, $\nu_n = \mathbb{E}(M_n)$ is such that $\frac{\nu_{n+1}}{c_n}$ converges to zero, for some α , $1 < \alpha \leq 2$, $\nu_{\alpha,n} = \mathbb{E}(\|M_n\|^\alpha) < \infty$ for all $n \geq 1$ and

$$\sum_{n=0}^{\infty} \frac{\eta_n^\alpha}{c_n^\alpha} \nu_{\alpha,n+1} < \infty. \quad (31)$$

(K5) $\{M_n : n \geq 1\}$ is a martingale difference sequence, for some α , $1 < \alpha \leq 2$, $\nu_{\alpha,n} = \mathbb{E}(\|M_n\|^\alpha) < \infty$ for all $n \geq 1$ and

$$\sum_{n=0}^{\infty} \frac{\eta_n^\alpha}{c_n^\alpha} \nu_{\alpha,n+1} < \infty. \quad (32)$$

5 Proofs

Here is an auxiliary result, which plays an important role in our approach. The second part of this can be thought of as an extension of Kronecker's lemma.

Lemma 7. Let $\{\tau_n : n \geq 0\} \subseteq (0, \infty)$ be such that

$$\sum_n \tau_n = \infty, \quad \sum_n \tau_n^2 < \infty. \quad (33)$$

Let $\{z_n : n \geq 1\} \subseteq \mathbb{R}^d$ and let $\{s_n : n \geq 0\} \subseteq \mathbb{R}^d$ be defined by $s_0 = 0$ and

$$s_{n+1} = (1 - \tau_n)s_n + \tau_n z_{n+1}. \quad (34)$$

Then we have

(i) If $\lim_n \|z_{n+1}\| = 0$, then $\lim_n \|s_n\| = 0$.

(ii) If $\sum_{k=0}^n \tau_k z_{k+1}$ converges (in $\mathbb{R}^d$), then also $\lim_n \|s_n\| = 0$.

Proof: Let us also note that in view of (33), $\exists n^*$ such that $\tau_j < 1$ for all $j \geq n^*$. We can check using induction on n that for $n^* \leq m \leq n$ we have

$$[\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] + \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k = 1, \quad (35)$$

$$[\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] s_m + \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k z_{k+1} = s_{n+1}. \quad (36)$$

Using (35) we can see that for $n^* \leq m \leq n$ we have

$$\sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k = 1 - [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] \leq 1. \quad (37)$$

If $\lim_n \|z_{n+1}\| = 0$ then given $\epsilon > 0$, we can choose $m > n^*$ such that $\|z_{n+1}\| < \epsilon$ for all $n \geq m$ and then for $n \geq m$ we have (in view of (37))

$$\begin{aligned} \|s_{n+1}\| &\leq [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] \|s_m\| + \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k \epsilon \\ &\leq [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] \|s_m\| + \epsilon. \end{aligned}$$

Taking limit as $n \rightarrow \infty$, we conclude that $\lim_n \|s_{n+1}\| \leq \epsilon$. Since this holds for all $\epsilon > 0$, we conclude that $\|s_n\| \rightarrow 0$ completing proof of (i). Coming to the proof of (ii), for $0 \leq m \leq n$, let

$$q_{m,n} = \sum_{j=m}^n \tau_j z_{j+1}, \quad (38)$$

In view of assumption that $\sum_{j=1}^n \tau_j z_{j+1}$ converges as $n \rightarrow \infty$, it follows that $\forall \epsilon > 0, \exists n_\epsilon$ such that

$$\|q_{m,n}\| \leq \epsilon \quad \forall n \geq m \geq n_\epsilon. \quad (39)$$

Now writing $c_{n+1,n} = 0$, it follows using (36) that

$$\begin{aligned} s_{n+1} &= [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] s_m + \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k z_{k+1} \\ &= [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] s_m + \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \tau_j)] (q_{k,n} - q_{k+1,n}) \\ &= [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] s_m + [\prod_{\{m < j \leq n\}} (1 - \tau_j)] q_{m,n} \\ &\quad + \sum_{t=m+1}^n [\prod_{\{t < j \leq n\}} (1 - \tau_j) - \prod_{\{t-1 < j \leq n\}} (1 - \tau_j)] q_{t,n} \end{aligned}$$

Thus, in view of (39), we have $\forall n \geq m \geq n_\epsilon$,

$$\begin{aligned} \|s_{n+1}\| &\leq [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] \|s_m\| + [\prod_{\{m < j \leq n\}} (1 - \tau_j)] \|q_{m,n}\| \\ &\quad + \sum_{t=m+1}^n [\prod_{\{t < j \leq n\}} (1 - \tau_j) - \prod_{\{t-1 < j \leq n\}} (1 - \tau_j)] \|q_{t,n}\| \\ &\leq [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] \|s_m\| + [\prod_{\{m < j \leq n\}} (1 - \tau_j)] \epsilon \\ &\quad + \sum_{t=m+1}^n [\prod_{\{t < j \leq n\}} (1 - \tau_j) - \prod_{\{t-1 < j \leq n\}} (1 - \tau_j)] \epsilon \\ &\leq [\prod_{\{m \leq j \leq n\}} (1 - \tau_j)] \|s_m\| + \epsilon. \end{aligned}$$

As seen in proof of (i), this yields $\|s_n\| \rightarrow 0$. □

We will need another observation, which is needed to consider the case of iid errors when the mean may not be defined.

Lemma 8. $\forall x > 0$

$$\frac{x}{(\log(1+x))^\delta} \leq y \implies \log(1+x) \leq 3 \log(1+y) \quad (40)$$

Proof: Let us note that for all $x > 0$, $(1+x) \leq \exp(\sqrt{2x})$ as can be easily seen. Hence $\log(1+x) \leq \sqrt{2x}$ and thus $\frac{x}{\log(1+x)} \geq \frac{x}{\sqrt{2x}} = \frac{\sqrt{x}}{\sqrt{2}}$.

If $\frac{x}{\log(1+x)} \leq y$ then $(1 + \frac{\sqrt{x}}{\sqrt{2}}) \leq (1+y)$. Since $(1 + \frac{\sqrt{x}}{\sqrt{2}})^3 \geq (1+x)$, we have $(1+x) \leq (1+y)^3$ from which the required conclusion follows by taking logarithm. This proves the case $\delta = 1$. If $\delta < 1$,

$$\frac{x}{(\log(1+x))^\delta} \geq \frac{x}{(\log(1+x))}$$

and hence the result follows from the case $\delta = 1$. □

We now come to the proof of Theorem 3.

Let us fix a $(\mathcal{F}_n)$ adapted sequence of random variables $\{\zeta_n : n \geq 1\}$ that is bounded, say by a constant C and let

$$S_{n+1} = (1 - t\beta_n)S_n + t\beta_n\zeta_n W_{n+1}, \quad n \geq 0.$$

Let us write $W_{n+1,i} = U_{n+1,i} + V_{n+1,i} + R_{n+1,i}$ where

$$\begin{aligned} U_{n+1,i} &= W_{n+1,i} \mathbf{1}_{A_{n+1,i}^c}, \\ V_{n+1,i} &= \mathbb{E}[W_{n+1,i} \mathbf{1}_{A_{n+1,i}} \mid \mathcal{F}_n], \\ R_{n+1,i} &= W_{n+1,i} \mathbf{1}_{A_{n+1,i}} - V_{n+1,i}. \end{aligned}$$

Let $H_0 = 0$, $J_0 = 0$ and $K_0 = 0$ for $n \geq 0$

$$\begin{aligned} H_{n+1,i} &= (1 - t\beta_n)H_n + t\beta_n\zeta_n U_{n+1,i} \\ J_{n+1,i} &= (1 - t\beta_n)J_n + t\beta_n\zeta_n V_{n+1,i} \\ K_{n+1,i} &= (1 - t\beta_n)K_n + t\beta_n\zeta_n R_{n+1,i}. \end{aligned}$$

Clearly, $S_{n+1,i} = H_{n+1,i} + J_{n+1,i} + K_{n+1,i}$ and we will prove that each of $H_{n+1,i}$, $J_{n+1,i}$ and $K_{n+1,i}$ converge to 0 almost surely as $n \rightarrow \infty$, $\forall i$. That will complete the proof of the theorem.

In view of condition (18) and Borel-Cantelli lemma, along with the fact that ζ_n is bounded, we have $\zeta_n U_{n+1,i}$ converges to zero almost surely, and hence in view of Lemma 7 it follows that $H_{n+1,i}$ converge to zero almost surely. If $V_{n+1,i}$ satisfies (22) then so does $\zeta_n V_{n+1,i}$ and as a consequence $J_{n+1,i}$ converges to 0 in view of (i) in Lemma 7 while if $V_{n+1,i}$ satisfies (19) then again so does $\zeta_n V_{n+1,i}$ and as a consequence $J_{n+1,i}$ converges to 0 in view of (ii) in Lemma 7. It is clear that

$$\begin{aligned} \mathbb{E}[|R_{n+1,i}|^2 \mid \mathcal{F}_n] &= \mathbb{E}[|W_{n+1,i}|^2 \mathbf{1}_{A_{n+1,i}} \mid \mathcal{F}_n] - |V_{n+1,i}|^2 \\ &\leq \mathbb{E}[|W_{n+1,i}|^2 \mathbf{1}_{A_{n+1,i}} \mid \mathcal{F}_n] \end{aligned}$$

and since $\{\zeta_n : n \geq 1\}$ is $(\mathcal{F}_n)$ adapted and is bounded (by C), we have

$$\mathbb{E}[|\zeta_n R_{n+1,i}|^2] \leq C^2 \mathbb{E}[|R_{n+1,i}|^2] \leq C^2 \mathbb{E}[|W_{n+1,i}|^2 \mathbf{1}_{A_{n+1,i}}]. \quad (41)$$

Thus $Q_{n+1,i} = \sum_{k=0}^n \beta_k \zeta_n R_{k+1,i}$ is an L^2 -bounded martingale in view of assumption (20) and thus

$$\sum_{k=0}^n \beta_k \zeta_n R_{k+1,i} \text{ converges almost surely.} \quad (42)$$

Invoking part (ii) of Lemma 7 it follows that $K_{n+1,i}$ converges to zero almost surely, completing the proof as remarked earlier. $\square$

We now proceed with the proof of Theorem 1. Let $\xi_n = \lambda_n(X_0, X_1, \dots, X_n)$ where λ_n satisfies (5) and let $\phi_n = (1 + \max_{0 \leq k \leq n} \|X_k\|)$. Let b be such that (7) holds and let $\zeta_n = -b^{-1}\phi_n^{-1}\xi_n$, $Z_{n+1} = \zeta_n W_{n+1}$, $\tau_n = b\beta_n$ and $R_n = b^{-1}G(X_n)$. Let us note that ζ_n is bounded by $b^{-1}C_1$. Since $\tau_n \phi_n Z_{n+1} = b\beta_n \phi_n \zeta_n W_{n+1} = -\beta_n \xi_n W_{n+1}$ and as a result we can rewrite (6) as

$$\begin{aligned} X_{n+1} &= X_n - \beta_n (G(X_n) + \xi_n W_{n+1}) \\ &= X_n - \tau_n b^{-1} G(X_n) - \beta_n \xi_n W_{n+1} \\ &= X_n - \tau_n R_n + \tau_n \phi_n Z_{n+1} \end{aligned}$$

and hence

$$X_{n+1} - x^* = (1 - \tau_n)(X_n - x^*) + \tau_n((X_n - x^*) - R_n) + \tau_n \phi_n Z_{n+1}. \quad (43)$$

Note that in view of the assumption (7) we have

$$\|((X_k - x^*) - R_k)\| \leq \rho \|X_k - x^*\|. \quad (44)$$

Before we proceed, let us note that ϕ_n is an increasing process. Since $\{W_n : n \geq 1\}$ satisfies GLLN at the rate $\{\beta_n : n \geq 0\}$, and ζ_n is bounded, $\tau_n = b\beta_n$ it follows that S_{n+1} converges to zero almost surely where $\{S_n : n \geq 0\}$ is defined by $S_0 = 0$ and

$$S_{n+1} = (1 - \tau_n)S_n + \tau_n Z_{n+1}, \quad n \geq 0. \quad (45)$$

We are going to show that for every $\omega \in \Omega$ such that $S_{n+1}(\omega)$ defined by (45) converges, $X_{n+1}(\omega) - x^*$ defined by (43) converges to zero. We will drop ω in the notation as usual, and we should keep in mind that choices of n, m, K in what follows can depend upon ω .

Let m^* be chosen such that $\tau_n = b\beta_n < 1$ for all $n \geq m^*$. Since $\{\beta_n : n \geq 0\}$ satisfies (8), so does $\{\tau_n : n \geq 0\}$. We can check, using recursion and (43) that for $0 \leq m \leq n$, we have

$$\begin{aligned} X_{n+1} - x^* &= \left[\prod_{j=m}^n (1 - \tau_j) \right] (X_m - x^*) \\ &\quad + \sum_{k=m}^n \left[\prod_{\{j:k < j \leq n\}} (1 - \tau_j) \right] \tau_k ((X_k - x^*) - R_k) \\ &\quad + \sum_{k=m}^n \left[\prod_{\{j:k < j \leq n\}} (1 - \tau_j) \right] \phi_k \tau_k Z_{k+1}. \end{aligned} \quad (46)$$

Here, product over an empty set is taken as 1. Let us introduce notation for $0 \leq m \leq n$,

$$\begin{aligned} U_{m,n} &= \left[\prod_{j=m}^n (1 - \tau_j) \right] (X_m - x^*) \\ V_{m,n} &= \sum_{k=m}^n \left[\prod_{\{j:k < j \leq n\}} (1 - \tau_j) \right] \tau_k ((X_k - x^*) - R_k) \\ L_{m,n} &= \sum_{k=m}^n \left[\prod_{\{k < j \leq n\}} (1 - \tau_j) \right] \phi_k \tau_k Z_{k+1} \\ D_{m,n} &= \sum_{k=m}^n \left[\prod_{\{k < j \leq n\}} (1 - \tau_j) \right] \tau_k Z_{k+1} \end{aligned}$$

Let us note (by recursion on n , as in (36)) that

$$S_{n+1} = \left[\prod_{\{m \leq j \leq n\}} (1 - \tau_j) \right] S_m + D_{m,n}. \quad (47)$$

In view of (46), we have for any $0 \leq m \leq n$

$$X_{n+1} - x^* = U_{m,n} + V_{m,n} + L_{m,n}$$

and hence (noting that $m^* \leq m \leq n$ implies $(1 - \tau_j) > 0$ for all $j, m \leq j \leq n$)

$$\|X_{n+1} - x^*\| \leq \|U_{m,n}\| + \|V_{m,n}\| + \|L_{m,n}\|, \quad \forall m^* \leq m \leq n. \quad (48)$$

Let $\gamma_n = \max\{\|X_k - x^*\|, \quad 0 \leq k \leq n\}$ and $\gamma^* = \sup_n \gamma_n$. Then $\|X_k\| \leq (\gamma_n + \|x^*\|)$ for $0 \leq k \leq n$, and recalling that $\phi_n = (1 + \max_{0 \leq k \leq n} \|X_k\|)$, it follows that

$$\phi_n \leq (1 + \gamma_n + \|x^*\|), \quad \forall n. \quad (49)$$

We will first prove that for all $\epsilon > 0$, $\exists n_\epsilon$ such that

$$\|L_{m,n}\| \leq (1 + \gamma_n + \|x^*\|)\epsilon \quad \forall n \geq m \geq n_\epsilon. \quad (50)$$

Recall, we are working with ω such that $S_{n+1} = S_{n+1}(\omega)$ converges to zero and hence given $\epsilon > 0$, we can choose $n_\epsilon > m^*$ such that

$$\|S_{n+1}\| \leq \frac{1}{2}\epsilon \quad \forall n \geq n_\epsilon.$$

Now in view of (47), it follows that

$$\|D_{m,n+1}\| \leq \epsilon \quad \forall n \geq m \geq n_\epsilon. \quad (51)$$

Let us note that

$$\begin{aligned} L_{m,n} &= \sum_{k=m}^n \phi_k [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k Z_{k+1} \\ &= \phi_m \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k Z_{k+1} \\ &\quad + \sum_{k=m+1}^n (\phi_k - \phi_m) [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k Z_{k+1} \\ &= \phi_m D_{m,n} + \sum_{k=m+1}^n (\phi_k - \phi_m) [\prod_{\{k < j \leq n\}} (1 - \tau_j)] \tau_k Z_{k+1} \end{aligned} \quad (52)$$

Proceeding likewise, we will get

$$L_{m,n} = \phi_m D_{m,n} + (\phi_{m+1} - \phi_m) D_{m+1,n} + \dots + (\phi_n - \phi_{n-1}) D_{n,n}$$

Since ϕ_k is increasing, in view of (51) it follows that $\forall n \geq m \geq n_\epsilon$,

$$\begin{aligned} \|L_{m,n}\| &\leq \phi_m \|D_{m,n}\| + (\phi_{m+1} - \phi_m) \|D_{m+1,n}\| + \dots + (\phi_n - \phi_{n-1}) \|D_{n,n}\| \\ &\leq \phi_m \epsilon + (\phi_{m+1} - \phi_m) \epsilon + \dots + (\phi_n - \phi_{n-1}) \epsilon \\ &= \phi_n \epsilon \\ &\leq (1 + \gamma_n + \|x^*\|) \epsilon \end{aligned}$$

where we have used (49) at the last step. This proves (50). Next we will prove that

$$\gamma^* = \sup\{\|X_n - x^*\| : n \geq 0\} < \infty. \quad (53)$$

First, using $\rho < 1$, get $\epsilon > 0$ such that $\epsilon < \frac{1}{3}$

$$\rho(1 + 3\epsilon) \leq 1. \quad (54)$$

For this ϵ , choose n_ϵ such that (50) holds. Now we will prove, by induction that $\forall k \geq n_\epsilon$

$$\gamma_k \leq (1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|). \quad (55)$$

Clearly (55) is true for $k = n_\epsilon$. Suppose that (55) holds for all k such that $n_\epsilon \leq k \leq n^*$. We will prove that (55) also holds for $k = n^* + 1$, completing the proof that (55) is true for all $k \geq n_\epsilon$. To prove the induction step, suffices to prove that

$$\|X_{n^*+1} - x^*\| \leq (1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|).$$

For $n_\epsilon \leq k \leq n^*$, (44), (54), (55) imply that

$$\begin{aligned} \|((X_k - x^*) - R_k)\| &\leq \rho \|X_k - x^*\| \\ &\leq \rho \gamma_k \\ &\leq \rho(1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|) \\ &\leq (1 + \gamma_{n_\epsilon} + \|x^*\|). \end{aligned} \quad (56)$$

As a consequence

$$\begin{aligned}\|V_{n_\epsilon, n^*}\| &\leq \sum_{k=n_\epsilon}^{n^*} [\prod_{\{j:k < j \leq n\}} (1 - \tau_j)] \tau_k \|((X_k - x^*) - R_k)\| \\ &\leq (1 + \gamma_{n_\epsilon} + \|x^*\|) \sum_{k=n_\epsilon}^{n^*} [\prod_{\{j:k < j \leq n^*\}} (1 - \tau_j)] \tau_k.\end{aligned}\tag{57}$$

It is easy to see that

$$\begin{aligned}\|U_{n_\epsilon, n^*}\| &= [\prod_{j=n_\epsilon}^{n^*} (1 - \tau_j)] \|X_{n_\epsilon} - x^*\| \\ &\leq \gamma_{n_\epsilon} [\prod_{j=n_\epsilon}^{n^*} (1 - \tau_j)].\end{aligned}\tag{58}$$

For all $m \leq n$ one can verify (by induction on n) that

$$[\prod_{k=m}^n (1 - \tau_k)] + \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \tau_j)] \tau_k = 1,\tag{59}$$

and then putting together (57), (58), (59) we conclude that

$$\|U_{n_\epsilon, n^*}\| + \|V_{n_\epsilon, n^*}\| \leq (1 + \gamma_{n_\epsilon} + \|x^*\|).\tag{60}$$

Using (50), along with (55) for $k = n^*$ we deduce

$$\begin{aligned}\|L_{n_\epsilon, n^*}\| &\leq \epsilon(1 + \gamma_{n^*} + \|x^*\|) \\ &\leq \epsilon(1 + \|x^*\|) + \epsilon\gamma_{n^*} \\ &\leq \epsilon(1 + \|x^*\|) + \epsilon(1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|) \\ &\leq (2\epsilon + 3\epsilon^2)(1 + \|x^*\|) + \epsilon(1 + 3\epsilon)\gamma_{n_\epsilon} \\ &\leq 3\epsilon(1 + \gamma_{n_\epsilon} + \|x^*\|).\end{aligned}\tag{61}$$

Here, since $\epsilon < \frac{1}{3}$, we have used $(2\epsilon + 3\epsilon^2) \leq 3\epsilon$ and also $(1 + 3\epsilon) \leq 2$. We thus conclude, using (48), (60) along with (61), that

$$\begin{aligned}\|X_{n^*+1} - x^*\| &\leq \|U_{n_\epsilon, n^*}\| + \|V_{n_\epsilon, n^*}\| + \|L_{n_\epsilon, n^*}\| \\ &\leq (1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|).\end{aligned}\tag{62}$$

This completes the induction proof and thus (55) holds for all $k \geq n_\epsilon$ proving (53).

It also follows that $\sup\{\|X_k\| : k \geq 0\} < \infty$ and hence that

$$\sup\{\phi_n : n \geq 0\} < \infty.\tag{63}$$

Using (50) and (63), it follows that for all $\delta > 0$, we can get m_δ such that

$$\|L_{m,n}\| < \delta \quad \forall m, n \text{ such that } m_\delta \leq m \leq n\tag{64}$$

and thus we have for $m \geq m_\delta$

$$\limsup_{n \rightarrow \infty} \|L_{m,n}\| \leq \delta.\tag{65}$$

Note that in view of the assumption that $\{\beta_n : n \geq 0\}$ satisfies (8) and $\tau_n = b\beta_n$, we have $\sum_n \tau_n = \infty$ and hence

$$\lim_{n \rightarrow \infty} [\prod_{j=m}^n (1 - \tau_j)] = 0, \quad \forall m < \infty.\tag{66}$$

For $m \geq 1$, let $\theta_m = \sup\{\|X_k - x^*\| : k \geq m\}$. Then clearly, θ_m is decreasing and $\theta_m \leq \gamma^* < \infty$ for all m . Let $\theta^* = \lim_{m \rightarrow \infty} \theta_m$. Clearly $\theta^* \leq \gamma^*$. Remains to show that $\theta^* = 0$. Since

$$\|U_{m,n}\| \leq \|X_m - x^*\| [\prod_{j=m}^n (1 - \tau_j)]$$

and thus in view of (66) we have

$$\lim_{n \rightarrow \infty} \|U_{m,n}\| = 0 \quad \forall m < \infty. \quad (67)$$

Note that for any $m \leq n$ we have

$$\begin{aligned} \|V_{m,n}\| &\leq \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \tau_j)] \tau_k \|((X_k - x^*) - R_k)\| \\ &\leq \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \tau_j)] \tau_k \rho \|X_k - x^*\| \\ &\leq \rho \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \tau_j)] \tau_k \theta_k \\ &\leq \rho \theta_m \end{aligned} \quad (68)$$

Hence for all m

$$\limsup_{n \rightarrow \infty} \|V_{m,n}\| \leq \rho \theta_m. \quad (69)$$

Combining (48) along with (65), (67) and (69), we get for any $m \geq m_\delta$

$$\limsup_{n \rightarrow \infty} \|X_{n+1} - x^*\| \leq \rho \theta_m + \delta.$$

Now taking limit as $m \rightarrow \infty$ on the RHS above and then limit as $\delta \downarrow 0$, we get

$$\limsup_{n \rightarrow \infty} \|X_{n+1} - x^*\| \leq \rho \theta^*.$$

From the definition of θ^* , it follows that $\limsup_{n \rightarrow \infty} \|X_{n+1} - x^*\| = \theta^*$ and hence we have shown

$$\theta^* \leq \rho \theta^*. \quad (70)$$

Since $\rho < 1$ and $0 \leq \theta^* < \infty$, (70) implies $\theta^* = 0$. This completes proof of Theorem 1. $\square$

Coming to the proof of Theorem 2, several steps are same as in the proof of Theorem 1. Recall that $\xi_n = \lambda_n(Y_0, Y_1, \dots, Y_n)$ where λ_n satisfies (12) and let $\phi_n = (1 + \max_{0 \leq k \leq n} \|Y_k\|_\infty)$. Let b be such that (11) holds and let $\zeta_n = -b^{-1}\phi_n^{-1}\xi_n$, $T_{n,i} = \frac{1}{2bc_n}(F(Y_n + e_i c_n) - F(Y_n - e_i c_n))$, $Z_{n+1,i} = \psi_{n,i}\zeta_n W_{n+1,i}$, $\tau_n = b\eta_n$. Then $\tau_n \phi_n Z_{n+1,i} = b\eta_n \psi_{n,i} \phi_n \zeta_n W_{n+1,i} = -\eta_n \psi_{n,i} \xi_n M_{n+1,i}$. Note that $\{\zeta_n : n \geq 0\}$ is bounded. For later use, let us note that in view of assumption (11) on F , it follows that

$$|(Y_{n,i} - x_i^*) - T_{n,i}| \leq \rho |Y_{n,i} - x_i^*| + ac_n. \quad (71)$$

Recalling that $W_{n+1} = \frac{1}{2c_n}(M'_{n+1} - M''_{n+1})$, $n \geq 0$, let us rewrite (15) as follows:

$$\begin{aligned} Y_{n+1} - x^* &= Y_n - x^* - \sum_i \psi_{n,i} \eta_n \left(\frac{(F(Y_n + e_i c_n) + \xi_n M'_{n+1,i}) - (F(Y_n - e_i c_n) + \xi_n M''_{n+1,i})}{2c_n} \right) e_i \\ &= Y_n - x^* - \sum_i \psi_{n,i} \tau_n \left(\frac{F(Y_n + e_i c_n) - F(Y_n - e_i c_n)}{2bc_n} \right) e_i - \sum_i \psi_{n,i} \eta_n \left(\frac{\xi_n (M'_{n+1,i} - M''_{n+1,i})}{2c_n} \right) e_i \\ &= \sum_i (1 - \psi_{n,i} \tau_n) (Y_{n,i} - x_i^*) e_i \\ &\quad + \sum_i \psi_{n,i} \tau_n (Y_{n,i} - x_i^* - \frac{1}{2bc_n} (F(Y_n + e_i c_n) - F(Y_n - e_i c_n))) e_i - \sum_i \psi_{n,i} \eta_n \xi_n W_{n+1,i} e_i \\ &= \sum_i ((1 - \psi_{n,i} \tau_n) (Y_{n,i} - x_i^*) + \psi_{n,i} \tau_n ((Y_{n,i} - x_i^*) - T_{n,i}) + \tau_n \phi_n Z_{n+1,i}) e_i \end{aligned}$$

and hence

$$Y_{n+1,i} - x_i^* = (1 - \psi_{n,i}\tau_n)((Y_{n,i} - x_i^*) + \psi_{n,i}\tau_n(Y_{n,i} - x_i^*) - T_{n,i}) + \tau_n\phi_n Z_{n+1,i}. \quad (72)$$

We are going to prove that for each i , $|Y_{n+1,i} - x_i^*|$ converges to zero.

Before we proceed, let us note that ϕ_n is an increasing process and also that when $\psi_{n,i} = 0$, $Z_{n+1,i} = 0$ and $Y_{n+1,i} - x_i^* = Y_{n,i} - x_i^*$. Since $\{W_n : n \geq 1\}$ satisfies GSKLN at the rate $\{\eta_n : n \geq 0\}$, and ζ_n is bounded, $\tau_n = b\eta_n$ it follows that $S_{n+1,i}$ converges to zero almost surely for each i , $1 \leq i \leq d$ where $\{S_{n,i} : n \geq 0\}$ is defined by $S_{0,i} = 0$ and

$$S_{n+1,i} = (1 - \tau_n)S_{n,i} + \tau_n Z_{n+1,i}, \quad n \geq 0. \quad (73)$$

Recall that $\{\eta_n, c_n, \psi_{n,i} : n \geq 0, 1 \leq i \leq d\}$ satisfy (13), (14). Thus $\sum_n \psi_{n,i}\tau_n = \infty$ almost surely. We are going to show that for every $\omega \in \Omega$ such that $S_{n+1,i}(\omega)$ defined by (73) converges and $\sum_n \psi_{n,i}(\omega)\tau_n = \infty$ for each i ; $Y_{n+1,i}(\omega) - x_i^*$ defined by (72) converges to zero, as in proof of Theorem 1. We will drop ω in the notation as usual, and we should keep in mind that choices of n, m, K in what follows can depend upon ω .

Let m^* be chosen such that $\tau_n = b\beta_n < 1$ for all $n \geq m^*$. Since $\{\beta_n : n \geq 0\}$ satisfies (8), so does $\{\tau_n : n \geq 0\}$. We can check, using recursion and (72) that for $0 \leq m \leq n$, we have

$$\begin{aligned} Y_{n+1,i} - x_i^* &= [\prod_{j=m}^n (1 - \psi_{j,i}\tau_j)(Y_{m,i} - x_i^*) \\ &\quad + \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \psi_{j,i}\tau_j)] \psi_{k,i}\tau_k T_{k,i} \\ &\quad + \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \psi_{j,i}\tau_j)] \psi_{n,i}\phi_k\tau_k Z_{k+1,i}. \end{aligned} \quad (74)$$

Here, product over an empty set is taken as 1. Let us introduce notation for $0 \leq m \leq n$,

$$U_{m,n,i} = [\prod_{j=m}^n (1 - \psi_{j,i}\tau_j)(Y_{m,i} - x_i^*)] \quad (75)$$

$$V_{m,n,i} = \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \psi_{j,i}\tau_j)] \psi_{k,i}\tau_k (Y_{m,i} - x_i^* - T_{k,i}) \quad (76)$$

$$L_{m,n,i} = \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \psi_{j,i}\tau_j)] \psi_{k,i}\phi_k\tau_k Z_{k+1,i} \quad (77)$$

$$D_{m,n,i} = \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \psi_{j,i}\tau_j)] \psi_{k,i}\tau_k Z_{k+1,i} \quad (78)$$

Let us note (by recursion on n , as in (36)) that

$$S_{n+1,i} = [\prod_{\{m \leq j \leq n\}} (1 - \psi_{j,i}\tau_j)] S_{m,i} + D_{m,n,i}. \quad (79)$$

In view of (74), we have for any $0 \leq m \leq n$

$$Y_{n+1,i} - x_i^* = U_{m,n,i} + V_{m,n,i} + L_{m,n,i}$$

and hence (noting that $m^* \leq m \leq n$ implies $(1 - \psi_{j,i}\tau_j) > 0$ for all j , $m \leq j \leq n$)

$$|Y_{n+1,i} - x_i^*| \leq |U_{m,n,i}| + |V_{m,n,i}| + |L_{m,n,i}|, \quad \forall m^* \leq m \leq n.$$

$$\|Y_{n+1} - x^*\|_\infty \leq \|U_{m,n}\|_\infty + \|V_{m,n}\|_\infty + \|L_{m,n}\|_\infty, \quad \forall m^* \leq m \leq n. \quad (80)$$

Let $\gamma_n = \max\{\|Y_k - x^*\|_\infty : 0 \leq k \leq n\}$ and $\gamma^* = \sup_n \gamma_n$. Note that for all i and $k \leq n$, $|Y_{k,i} - x_i^*| \leq \gamma_n$. By definition, we have $\|Y_n\|_\infty \leq (\gamma_n + \|x^*\|_\infty)$ and recalling that $\phi_n = (1 + \max_{0 \leq k \leq n} \|Y_k\|_\infty)$, it follows that

$$\phi_n \leq (1 + \gamma_n + \|x^*\|_\infty) \quad \forall n. \quad (81)$$

We will first prove that for all $\epsilon > 0$, $\exists n_\epsilon$ such that

$$|L_{m,n,i}| \leq (1 + \gamma_n + \|x^*\|_\infty)\epsilon \quad \forall n \geq m \geq n_\epsilon, 1 \leq i \leq d. \quad (82)$$

Recall, we are working with ω such that $S_{n+1} = S_{n+1}(\omega)$ converges to zero and hence given $\epsilon > 0$, we can choose $n_\epsilon > m^*$ such that

$$\|S_{n+1}\|_\infty \leq \frac{1}{2}\epsilon \quad \forall n \geq n_\epsilon$$

and thus $|S_{n+1,i}| \leq \frac{1}{2}\epsilon \quad \forall n \geq n_\epsilon$. In view of (79), it follows that

$$|D_{m,n+1,i}| \leq \epsilon \quad \forall n \geq m \geq n_\epsilon. \quad (83)$$

As seen in proof of Theorem 1, it follows that

$$\begin{aligned} L_{m,n,i} &= \sum_{k=m}^n \phi_k [\prod_{\{k < j \leq n\}} (1 - \psi_{j,i} \tau_j)] \psi_{k,i} \tau_k Z_{k+1,i} \\ &= \phi_m \sum_{k=m}^n [\prod_{\{k < j \leq n\}} (1 - \psi_{j,i} \tau_j)] \psi_{k,i} \tau_k Z_{k+1,i} \\ &\quad + \sum_{k=m+1}^n (\phi_k - \phi_m) [\prod_{\{k < j \leq n\}} (1 - \psi_{j,i} \tau_j)] \psi_{k,i} \tau_k Z_{k+1,i} \\ &= \phi_m D_{m,n} + \sum_{k=m+1}^n (\phi_k - \phi_m) [\prod_{\{k < j \leq n\}} (1 - \psi_{j,i} \tau_j)] \psi_{k,i} \tau_k Z_{k+1,i} \end{aligned} \quad (84)$$

Proceeding likewise, we will get

$$L_{m,n,i} = \phi_m D_{m,n} + (\phi_{m+1} - \phi_m) D_{m+1,n,i} + \dots + (\phi_n - \phi_{n-1}) D_{n,n,i}$$

Since ϕ_k is increasing, in view of (83) it follows that $\forall n \geq m \geq n_\epsilon$,

$$\begin{aligned} |L_{m,n,i}| &\leq \phi_m |D_{m,n,i}| + (\phi_{m+1} - \phi_m) |D_{m+1,n,i}| + \dots + (\phi_n - \phi_{n-1}) |D_{n,n,i}| \\ &\leq \phi_m \epsilon + (\phi_{m+1} - \phi_m) \epsilon + \dots + (\phi_n - \phi_{n-1}) \epsilon \\ &= \phi_n \epsilon \\ &\leq (1 + \gamma_n + \|x^*\|_\infty) \epsilon \end{aligned}$$

where we have used (81) at the last step. This proves (82). Next we will prove that

$$\gamma^* = \sup\{\|Y_n - x^*\|_\infty : n \geq 0\} < \infty. \quad (85)$$

First, using $\rho < 1$, get $\epsilon > 0$ such that $\epsilon < \frac{1}{3}$

$$\rho(1 + 3\epsilon) + \epsilon \leq 1. \quad (86)$$

For this ϵ , choose n_ϵ such that (82) holds and also

$$ac_n \leq \epsilon \quad \forall n \geq n_\epsilon, \quad (87)$$

with a such that (11) holds. This can be done since c_n converges to zero in view of assumption which can be done since $\{c_n : n \geq 1\}$ satisfies (13).

Now we will prove, by induction on k that $\forall k \geq n_\epsilon$

$$\gamma_k \leq (1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|_\infty). \quad (88)$$

Clearly (88) is true for $k = n_\epsilon$. Suppose that (88) holds for all k such that $n_\epsilon \leq k \leq n^*$. We will prove that (88) also holds for $k = n^* + 1$, completing the proof that (88) is true for all $k \geq n_\epsilon$. To prove the induction step, suffices to prove that for all $1 \leq i \leq d$

$$|Y_{n^*+1,i} - x_i^*| \leq (1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x_i^*\|_\infty). \quad (89)$$

This follows trivially of $\psi_{n^*,i} = 0$. When $\psi_{n^*,i} = 1$ For $n_\epsilon \leq k \leq n^*$, as noted in (71)

$$\begin{aligned} |Y_{k,i} - x_i^* - T_{k,i}| &\leq \rho |Y_{k,i} - x_i^*| + ac_n \\ &\leq \rho \gamma_k + ac_n \\ &\leq \rho(1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|_\infty) + \epsilon \\ &\leq (1 + \gamma_{n_\epsilon} + \|x^*\|_\infty) \end{aligned} \quad (90)$$

where we have used (85) and (87). As a consequence

$$\begin{aligned} |V_{n_\epsilon, n^*, i}| &\leq \sum_{k=n_\epsilon}^{n^*} [\prod_{\{j:k < j \leq n\}} (1 - \psi_{j,i} \tau_j)] \psi_{k,i} \tau_k |T_{k,i}| \\ &\leq (1 + \gamma_{n_\epsilon} + \|x^*\|_\infty) \sum_{k=n_\epsilon}^{n^*} [\prod_{\{j:k < j \leq n^*\}} (1 - \psi_{j,i} \tau_j)] \psi_{k,i} \tau_k. \end{aligned} \quad (91)$$

It is easy to see that

$$\begin{aligned} |U_{n_\epsilon, n^*, i}| &= [\prod_{j=n_\epsilon}^{n^*} (1 - \psi_{j,i} \tau_j)] |Y_{n_\epsilon, i} - x_i^*| \\ &\leq \gamma_{n_\epsilon} [\prod_{j=n_\epsilon}^{n^*} (1 - \psi_{j,i} \tau_j)]. \end{aligned} \quad (92)$$

For all $m \leq n$ and $1 \leq i \leq d$ one can verify (by induction on n) that

$$\left([\prod_{k=m}^n (1 - \psi_{k,i} \tau_k)] + \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \psi_{j,i} \tau_j)] \psi_{k,i} \tau_k \right) = 1, \quad (93)$$

and then putting together (91), (92), (93) we conclude that

$$|U_{n_\epsilon, n^*, i}| + |V_{n_\epsilon, n^*, i}| \leq (1 + \gamma_{n_\epsilon} + \|x^*\|_\infty). \quad (94)$$

Using (82), along with (88) for $k = n^*$ we deduce

$$\begin{aligned} |L_{n_\epsilon, n^*, i}| &\leq \epsilon(1 + \gamma_{n^*} + \|x^*\|_\infty) \\ &\leq \epsilon(1 + \|x^*\|_\infty) + \epsilon \gamma_{n^*} \\ &\leq \epsilon(1 + \|x^*\|_\infty) + \epsilon(1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|_\infty) \\ &\leq (2\epsilon + 3\epsilon^2)(1 + \|x^*\|_\infty) + \epsilon(1 + 3\epsilon)\gamma_{n_\epsilon} \\ &\leq 3\epsilon(1 + \gamma_{n_\epsilon} + \|x^*\|_\infty). \end{aligned} \quad (95)$$

Here, since $\epsilon < \frac{1}{3}$, we have used $(2\epsilon + 3\epsilon^2) \leq 3\epsilon$ and also $(1 + 3\epsilon) \leq 2$. We thus conclude, using (80), (94) along with (95), that

$$\begin{aligned} |Y_{n^*+1,i} - x_i^*| &\leq |U_{n_\epsilon, n^*, i}| + |V_{n_\epsilon, n^*, i}| + |L_{n_\epsilon, n^*, i}| \\ &\leq (1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|_\infty). \end{aligned} \quad (96)$$

Thus

$$\|Y_{n^*+1} - x^*\|_\infty \leq (1 + 3\epsilon)(1 + \gamma_{n_\epsilon} + \|x^*\|_\infty).$$

This completes the induction proof and thus (88) holds for all $k \geq n_\epsilon$ proving (85).

It also follows that $\sup\{\|Y_k\|_\infty : k \geq 0\} < \infty$ and hence that

$$\sup\{\phi_n : n \geq 0\} < \infty. \quad (97)$$

Using (82) and (97), it follows that for all $\delta > 0$, we can get m_δ such that

$$|L_{m,n,i}| < \delta \quad \forall m, n \text{ such that } m_\delta \leq m \leq n, \forall 1 \leq i \leq d \quad (98)$$

and thus we have for $m \geq m_\delta$, $1 \leq i \leq d$

$$\limsup_{n \rightarrow \infty} |L_{m,n,i}| \leq \delta \quad \forall i$$

and hence for all $m \geq m_\delta$

$$\limsup_{n \rightarrow \infty} \|L_{m,n}\|_\infty \leq \delta. \quad (99)$$

Since $\sum_n \psi_{n,i} \tau_n = \infty$ we have

$$\lim_{n \rightarrow \infty} [\prod_{j=m}^n (1 - \psi_{j,i} \tau_j)] = 0, \quad \forall m < \infty. \quad (100)$$

For $m \geq 1$, let $\theta_m = \sup\{\|Y_k - x^*\|_\infty : k \geq m\}$. Then clearly, θ_m is decreasing and $\theta_m \leq \gamma^* < \infty$ for all m . Let $\theta^* = \lim_{m \rightarrow \infty} \theta_m$. Clearly $\theta^* \leq \gamma^*$. Remains to show that $\theta^* = 0$. Since

$$|U_{m,n,i}| \leq |Y_{m,i} - x_i^*| [\prod_{j=m}^n (1 - \psi_{j,i} \tau_j)]$$

and thus in view of (100) we have

$$\lim_{n \rightarrow \infty} \|U_{m,n}\|_\infty = 0 \quad \forall m < \infty. \quad (101)$$

Note that for any $m \leq n$ we have

$$\begin{aligned} |V_{m,n,i}| &\leq \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \psi_{j,i} \tau_j)] \psi_{k,i} \tau_k \rho |Y_{k,i} - x_i^*| \\ &\leq \sum_{k=m}^n [\prod_{\{j:k < j \leq n\}} (1 - \psi_{j,i} \tau_j)] \tau_k \rho \theta_m \\ &\leq \rho \theta_m \end{aligned} \quad (102)$$

Hence for all m , $\limsup_{n \rightarrow \infty} |V_{m,n,i}| \leq \rho \theta_m$ and as a consequence

$$\limsup_{n \rightarrow \infty} \|V_{m,n}\|_\infty \leq \rho \theta_m. \quad (103)$$

Combining (80) along with (99), (101) and (103), we get for any $m \geq m_\delta$

$$\limsup_{n \rightarrow \infty} \|Y_{n+1} - x^*\|_\infty \leq \rho \theta_m + \delta.$$

Now taking limit as $m \rightarrow \infty$ on the RHS above and then limit as $\delta \downarrow 0$, we get

$$\limsup_{n \rightarrow \infty} \|Y_{n+1} - x^*\|_\infty \leq \rho \theta^*.$$

From the definition of θ^* , it follows that $\limsup_{n \rightarrow \infty} \|Y_{n+1} - x^*\|_\infty = \theta^*$ and hence we have shown

$$\theta^* \leq \rho\theta^*. \quad (104)$$

Since $\rho < 1$ and $0 \leq \theta^* \leq \gamma^* < \infty$, (104) implies that $\theta^* = 0$. This completes the proof of Theorem 2. $\square$

Coming to the proof of Theorem 5, we start with noting that under conditions (H1) as well as the case $\alpha = 2$ under (H5), the conditions of Theorem 3 are satisfied with $A_{n,i}^c = \phi$ (the empty set)- condition (18) holds trivially and (22) holds because of the condition on $\{W_n\}$ that $E(W_n) = 0$ under (H1) and $\{W_n\}$ being a martingale difference sequence under (H5). Likewise, (20) holds in view of (23) under (H1) and (27) under (H5).

Let us now come to (H5) for any $\alpha \in (1, 2)$. Let us define $A_{n,i} = \{\beta_n |W_{n,i}| \leq 1\}$. Now $P(A_{n,i}^c) = P(\beta_n^\alpha |W_{n,i}|^\alpha > 1)$ and thus $P(A_{n,i}^c) \leq \beta_n^\alpha E(|W_{n+1,i}|^\alpha) \leq \beta_n^\alpha \nu_{\alpha,n+1}$. Hence $\sum_n P(A_{n,i}^c) < \infty$ in view of (27) showing that (18) holds in this case. Since $\{W_n : n \geq 1\}$ is a martingale difference sequence, we have

$$V_{n+1,i} = E[W_{n+1,i} \mathbf{1}_{A_{n+1,i}} \mid \mathcal{F}_n] = -E[W_{n+1,i} \mathbf{1}_{A_{n+1,i}^c} \mid \mathcal{F}_n]$$

and hence

$$\begin{aligned} \beta_n E[|V_{n+1,i}|] &= \beta_n E[|E[W_{n+1,i} \mathbf{1}_{A_{n+1,i}^c} \mid \mathcal{F}_n]|] \\ &\leq \beta_n E[E[|W_{n+1,i}| \mathbf{1}_{A_{n+1,i}^c} \mid \mathcal{F}_n]] \\ &\leq E[\beta_n |W_{n+1,i}| \mathbf{1}_{A_{n+1,i}^c}] \\ &\leq E[\beta_n^\alpha |W_{n+1,i}|^\alpha \mathbf{1}_{A_{n+1,i}^c}] \end{aligned}$$

since $a^\alpha > a$ when $a > 1$ and $1 < \alpha$. Thus

$$E[\sum_n \beta_n |V_{n+1,i}|] \leq E[\sum_n \beta_n^\alpha |W_{n+1,i}|^\alpha] < \infty$$

in view of the assumption (27). Thus (19) holds. Coming to the third condition, using $a^\alpha \geq a^2$ when $0 < a \leq 1$ and $1 < \alpha < 2$, we have

$$\begin{aligned} \sum_n \beta_n^2 E(|V_{n+1,i}|^2) &\leq \sum_n \beta_n^\alpha E(|W_{n+1,i}|^\alpha \mathbf{1}_{A_{n+1,i}}) \\ &\leq \sum_n \beta_n^\alpha E(|W_{n+1,i}|^\alpha) \end{aligned}$$

and thus (27) implies (20). Thus GLLN holds if (H5) holds. Coming to (H4), it is easy to see that $U_n = W_n - E(W_n)$ is a martingale difference sequence and that $\{U_n : n \geq 0\}$ satisfies (H5) (with the same α). Thus $\{U_n : n \geq 0\}$ satisfies GLLN for the chosen step size $\{\beta_n : n \geq 0\}$. GLLN works $\mu_n = E(W_n)$ for the chosen step size does since μ_n converges to 0, see part (i) of Lemma 7. Thus GLLN holds under (H4).

Let us now look at (H2). This includes the case $\alpha = 1$ while in (H5), we need $1 < \alpha \leq 2$. Moreover, $\beta_n = n^{-\frac{1}{\alpha}}$ satisfies (L2) but not (L5).

Let us define $A_{n+1,i} = \{|W_{n+1,i}|^\alpha \leq n\}$. Since $\{W_n : n \geq 1\}$ are i.i.d. we have

$$P(A_{n+1,i}^c) = P(|W_{n+1,i}|^\alpha > n) = P(|W_{1,i}|^\alpha > n).$$

Since $\mathbb{E}(|W_{1,i}|^\alpha) < \infty$, it follows that $\sum_n \mathbb{P}(|W_{1,i}|^\alpha > n) < \infty$ and hence (18) holds.

Since $\{W_n : n \geq 1\}$ are i.i.d. and $\mathbb{E}[W_{1,i}] = 0$, we have

$$V_{n+1,i} = \mathbb{E}[W_{n+1,i} \mathbf{1}_{A_{n+1,i}} \mid \mathcal{F}_n] = \mathbb{E}[W_{n+1,i} \mathbf{1}_{A_{n+1,i}}] = \mathbb{E}[W_{1,i} \mathbf{1}_{\{|W_{1,i}|^\alpha \leq n\}}] = -\mathbb{E}[W_{1,i} \mathbf{1}_{\{|W_{1,i}|^\alpha > n\}}].$$

Thus

$$|V_{n+1,i}| \leq \mathbb{E}[|W_{1,i}| \mathbf{1}_{\{|W_{1,i}|^\alpha > n\}}] \leq \mathbb{E}[|W_{1,i}|^\alpha \mathbf{1}_{\{|W_{1,i}|^\alpha > n\}}].$$

Since $\mathbb{E}[|W_{1,i}|^\alpha] < \infty$ and we conclude $|V_{n+1,i}|$ converges to zero and hence (22) holds.

Coming to the (20), we have (recall that $1 \leq \alpha < 2$)

$$\begin{aligned} \sum_{n=0}^{\infty} \beta_n^2 \mathbb{E}[|W_{n+1,i}|^2 \mathbf{1}_{A_{n+1,i}}] &\leq C_3 + \sum_{n=2}^{\infty} D^2 n^{-\frac{2}{\alpha}} \mathbb{E}[|W_{1,i}|^2 \mathbf{1}_{\{|W_{1,i}|^\alpha \leq n\}}] \\ &= C_3 + D^2 \mathbb{E}[|W_{1,i}|^2 \sum_{n=2}^{\infty} n^{-\frac{2}{\alpha}} \mathbf{1}_{\{|W_{1,i}|^\alpha \leq n\}}] \end{aligned}$$

Noting that for $\theta > 1$ and $T > 0$ we have $\sum_{n=2}^{\infty} n^{-\theta} \mathbf{1}_{\{T \leq n\}} \leq 2T^{1-\theta}$ we can conclude that

$$\begin{aligned} \sum_{n=0}^{\infty} \beta_n^2 \mathbb{E}[|W_{n+1,i}|^2 \mathbf{1}_{A_{n+1,i}}] &\leq C_3 + 2D^2 \mathbb{E}[(|W_{1,i}|^2 (|W_{1,i}|^\alpha)^{(1-\frac{2}{\alpha}})] \\ &= C_3 + 2D^2 \mathbb{E}[(|W_{1,i}|^\alpha)] \\ &< \infty \end{aligned}$$

in view of the assumption that $\mathbb{E}(\|W_1\|^\alpha) < \infty$ and hence (20) follows.

Remains to show that conclusion is valid if (H3) holds. Here we define, for $n \geq 1$ $A_{n,i} = \{\frac{|W_{n,i}|}{(\log(1+|W_{n,i}|))^\delta} \leq n\}$. Since $\mathbb{E}[\frac{|W_{n,i}|}{(\log(1+|W_{n,i}|))^\delta}] < \infty$ for each i , it follows that $\sum_i \mathbb{P}(A_{n+1,i}) < \infty$ and hence (18) holds. Since $\{W_n : n \geq 1\}$ are i.i.d. with symmetric distribution (*i.e.* W_1 and $-W_1$ have the same distribution) it follows that $W_{n,i} \mathbf{1}_{A_{n+1,i}}$ has symmetric distribution for each i and hence (19) holds trivially. Coming to (20), remains to show that

$$\sum_{n=0}^{\infty} \beta_n^2 \mathbb{E}[|W_{n+1,i}|^2 \mathbf{1}_{A_{n+1,i}}] < \infty$$

In view of (25), suffices to prove that

$$\sum_{n=3}^{\infty} \frac{1}{n^2 (\log(1+n))^{2\delta}} \mathbb{E}[|W_{1,i}|^2 \mathbf{1}_{\{\frac{|W_{1,i}|}{(\log(1+|W_{1,i}|))^\delta} \leq n\}}] < \infty. \quad (105)$$

As seen in Lemma 8, on the set $\{\frac{|W_{1,i}|}{(\log(1+|W_{1,i}|))^\delta} \leq n\}$, $\log(1+|W_{1,i}|) \leq 3 \log(1+n)$ and thus

$$\frac{1}{(\log(1+|W_{1,i}|))^{2\delta}} \geq \frac{1}{9(\log(1+n))^{2\delta}}.$$

Hence to prove (105) suffices to prove that

$$\sum_{n=3}^{\infty} \frac{1}{n^2} \mathbb{E}[\frac{|W_{1,i}|^2}{(\log(1+|W_{1,i}|))^{2\delta}} \mathbf{1}_{\{\frac{|W_{1,i}|}{(\log(1+|W_{1,i}|))^\delta} \leq n\}}] < \infty. \quad (106)$$

Since $\frac{|W_{1,i}|}{(\log(1+|W_{1,i}|))^\delta}$ is integrable, (106) follows as seen above in the proof under condition (H2) for $\alpha = 1$. This completes the proof of Theorem 5

Let us now come to Theorem 6. Let us write $\theta_n = \frac{\eta_n}{c_n}$. Following steps used in the proof of Theorem 5, we can assert that $\{M_n : n \geq 1\}$ satisfies GSSLN at the rate $\{\theta_n : n \geq 1\}$, under each of the conditions (K1) - (K5). Note that here, the distribution of M_n is symmetric and the choice of $A_{n,i}$ ensures that V_n defined by (17) satisfies $V_n = 0$. Thus (19) trivially holds. Thus Theorem implies that

$$\sum_{n=0}^{\infty} M_{n+1} \theta_n \text{ converges almost surely.}$$

Noting that $W_{n+1} \eta_n = M_{n+1} \theta_n$, we conclude that

$$\sum_{n=0}^{\infty} W_{n+1} \eta_n \text{ converges almost surely}$$

and hence $\{W_n : n \geq 1\}$ satisfies GSSLN at the rate $\{\eta_n : n \geq 1\}$. As a consequence, if in addition, (11), (12), (13), (14) are satisfied, Stochastic Gradient Descent Theorem 2 holds. $\square$

6 Comments:

Thus we have shown that GSSLN works for $\beta_n = \frac{1}{n}$ if the errors $\{W_n\}$ are

(i) i.i.d. with mean 0 and $E(\|W_1\|) < \infty$

or

(ii) are independent with $\lim_n E(W_n) = 0$ and $\sup_n [E(\|W_n\|^\alpha)] < \infty$ for some $\alpha \in (1, 2]$.

or

(iii) are a martingale difference sequence with $\sup_n [E(\|W_n\|^\alpha)] < \infty$ for some $\alpha > 1$.

Thus Stochastic approximation works in all these cases for step size $\beta_n = \frac{1}{n}$. For Stochastic Gradient Descent, the same applies for step size $\eta_n = \frac{1}{n \log(1+n)}$ and $c_n = \frac{1}{\log(1+n)}$ for $n \geq 1$.

In the special case where $\{W_n\}$ are iid with symmetric distribution (*i.e.* W_1 and $-W_1$ have the same distribution), $E[\frac{|W_{1,i}|}{\log(1+|W_{1,i}|)^\delta}] < \infty$ for each i , (for some $0 \leq \delta \leq 1$) then GSSLN holds and so does Stochastic approximation if $\beta_n = \frac{1}{n \log(1+n)^\delta}$, $\forall n \geq 1$.

$E[\frac{|W_{1,i}|}{\log(1+|W_{1,i}|)^\delta}] < \infty$ with $\delta < 1$, then it can be seen that Stochastic Gradient Descent will hold for step size $\eta_n = \frac{1}{n \log(1+n)}$ and $c_n = \frac{1}{\log(1+n)^{1-\delta}}$ for $n \geq 1$.

Of course in each case discussed above, if supremum of higher order moment is finite then required above, one can choose appropriate β_n (and c_n for Stochastic Gradient Descent) so as to improve rate of convergence.

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